

RACHAD LAKIS

AI Consultant & ML Engineer | Computer Vision · NLP · Generative AI

Lebanon · rachadlakis@gmail.com · +961 3 962 918 · [LinkedIn](#) · [GitHub](#) · [Portfolio](#)

PROFESSIONAL SUMMARY

ML Engineer with 3+ years of experience and dual Master's degrees (Data Science, Computer Science), backed by two IEEE publications in Computer Vision and NLP. Builds across the full AI stack — from NumPy-level backpropagation to multi-GPU distributed training (DDP/FSDP) — spanning object detection, LLM fine-tuning, RAG pipelines, MCP integrations, and multi-agent systems. Currently consulting on AI architecture and engineering for clients across Lebanon and the UAE, including production agentic platforms and computer vision projects.

CORE EXPERTISE

- **Computer Vision:** YOLO, RF-DETR, object detection, image classification, dataset engineering
- **NLP & LLMs:** fine-tuning (LoRA/QLoRA, PEFT, bitsandbytes), prompt engineering, RAG, MCP
- **Agentic AI:** multi-agent orchestration, A2A protocol, Google ADK, tool calling
- **Distributed Training:** DDP, FSDP, ZeRO, mixed precision (bfloat16), gradient checkpointing, layer prefetching
- **Frameworks:** PyTorch, TensorFlow, Hugging Face, Ultralytics, Scikit-learn, OpenCV
- **Engineering & Infra:** FastAPI, PostgreSQL, AWS SageMaker/S3, n8n

WORK EXPERIENCE

AI Consultant — [devtech.pro](#) Jun 2023 – Present | Lebanon

- Deliver end-to-end ML/AI projects for clients spanning computer vision (image classification, object detection, motion detection) and generative AI, using PyTorch.
- Design, build, and deploy LLM-powered chatbots with RAG pipelines (chunking, embedding, Qdrant vector storage) and function calling / MCP integrations to connect models with external APIs and databases.
- Lead end-to-end data science workflows: collection, cleaning, preprocessing, analysis, and visualization.
- Build AI-driven automation for content creation, social publishing, and marketing workflows using low-code platforms (n8n).
- Designed AI agent architectures and delivered a full AI curriculum for a government digital-transformation client in the UAE (Abu Dhabi), covering AI fundamentals through production-grade applications.

Machine Learning Engineer — [Robots Go Mental](#) Jul 2024 – Oct 2024

- Implemented Guided Transfer Learning (GTL) in pure PyTorch to improve neural network training efficiency on limited data, without specialized fine-tuning libraries.
- Fine-tuned LLaMA 3.2-1B and BERT and ran comparative experiments evaluating performance with and without GTL.
- Used AWS SageMaker for large-scale training and S3 for experiment tracking and result storage.

AI Developer (Ideation) — [mightclick.ai](#) Jul 2024 – Oct 2024 | Remote

- Conceptualized and designed the AI components of a Chrome extension enabling text/comment generation, translation, summarization, audio, and image understanding via a right-click interface.
- Collaborated on product ideation and AI feature design ahead of public release.

AI Solutions Architect — [OMNIPEAK](#) Mar 2024 – Jun 2024

- Designed AI-driven agentic workflows for B2B applications, integrating LLMs with APIs and CRM platforms including HubSpot.
- Rapidly prototyped on low-code platforms (n8n, Node-RED) before production implementation in pure Python.

Machine Learning Engineer (Internship) — [AI Directions](#) 2023 | Dubai, UAE

- Collected, trained, and evaluated models for age detection using transfer learning and fine-tuning.
- Worked with CNN architectures (VGG-19, MobileNet-V2, ResNet-50) for facial and non-facial age estimation, integrating pose-estimation key points to improve classification accuracy.

Teaching Assistant — [Lebanese International University](#) Aug 2017 – Jun 2018

- Assisted in teaching Intermediate Java and Object-Oriented Programming during Master's studies; graded assignments and supported students with debugging.

Information Technology System Administrator — AMHHS Aug 2016 – May 2021 | Lebanon

- Managed the school system, helped with planning, accounting, and online teaching.

SELECTED PROJECTS

Augur — Multi-Agent Stock Trading System

- Applied the same multi-agent orchestration pattern (A2A, orchestrator + specialized agents) to build an automated stock trading system, with agents coordinating market analysis and trade decisions.

Mobius Tensors — Agentic LLM Training Orchestration

- Extended OMNI-Train with an agentic layer, using coordinated agents to automate and manage distributed LLM training pipelines.

Octopus-AI — Multi-Agent Orchestration Platform

- Built a modular platform (Google ADK, A2A protocol) with a central orchestrator coordinating five specialized agents for document services, planning, risk monitoring, and reporting; implemented Sequential and Loop agent patterns and exposed each agent as an independent A2A HTTP service.

github.com/rachadlakis/octopus-ai

OMNI-Train — Distributed Training Framework

- Built a multi-GPU training framework (CLI + web app) supporting DDP and FSDP for any PyTorch/Hugging Face model, with mixed precision, gradient checkpointing, layer prefetching, PEFT, and quantization; validated at scale training across 8 GPUs (RunPod).

github.com/rachadlakis/omni-train

Agentic-AI Framework — Full-Stack Gen AI Platform

- Built a full-stack agentic platform from scratch (FastAPI, PostgreSQL, PydanticAI) supporting multi-agent orchestration, tool calling, RAG, MCP integration, Celery task scheduling, and real-time voice chat (STT/TTS).

github.com/rachadlakis/agentic-ai-framework

Argus — Fire, Smoke & Threat Detection (Computer Vision)

- Curated a 120,000+ image dataset (merged COCO/YOLO formats) and trained/evaluated multiple YOLO variants (n/s/m/l/xl) and RF-DETR models for real-time fire, smoke, and firearm detection.

MCP Client & Server — Fleet Management Integration

- Implemented Model Context Protocol client and server in Python for Cooriroo, a fleet management system.

MNIST Digit Recognition — From Scratch

- Built a digit-recognition model using pure Python and NumPy — implementing backpropagation and gradient descent manually, without PyTorch or TensorFlow.

EDUCATION

Saint Joseph University of Beirut (USJ) — Master's in Data Science 2021 – 2023

Average 16.5/20 | Thesis: Pose Estimation Key Points in Age Recognition of Full Body Images (18/20, published)

Lebanese International University (LIU) — Master's in Computer Science 2016 – 2018

GPA 3.8/4.0 | Thesis: Improved TFIDF Weighting Techniques in Document Retrieval (Grade A, published)

Lebanese International University (LIU) — BS in Computer Science 2008 – 2013

Lebanese University, Faculty of Science — Pure Mathematics 2006 – 2008

PUBLICATIONS & AWARDS

- Lakis, R., Constantin, J., Constantin, I., Hoang, V.T., Ruichek, Y. — “Pose Estimation Keypoints in Age Recognition of Full Body Image,” IEEE IMCET, 2023
- Yamout, F., Lakis, R. — “Improved TFIDF Weighting Techniques in Document Retrieval,” IEEE ICDIM, 2018
- 3rd Place, Klango Fintech Award (ZAKA), 2022 — international NLP competition; built a Twitter sentiment-analysis system for financial topics

LANGUAGES

Arabic: Native | English: Fluent | French: Intermediate